MIDTERM STUDY GUIDE

Computer Security · Weeks 1–3, 5 and 6

Condensed, comprehensive review for efficient midterm preparation

What is valuable?	What can harm it?	Why could it succeed?	How serious?	What do we do?

Scope note

This guide covers every lecture in the repo: weeks 1, 2, 3, 5 and 6. **There are no week 4 slides in the folder.** If a week 4 lecture was given, get those slides before the exam, because nothing here covers it.

Important exam note

The slides never state the midterm format, so this guide prepares for the three things every lecture practised: (1) **define a term** in plain English, (2) **classify a scenario** (which CIA goal, which malware type, which phishing label, which risk level), and (3) **explain a real case** using asset / threat / vulnerability / control. The course teaches protection, not attack: no question will ask you how to perform an attack.

Prepared from the week 1–6 slides, my notes and the assignments.

How to Use This Guide

This guide replaces re-reading five slide decks. It keeps the lecture order, the lecturer's own terms and every real case, and drops repeated examples.

Study in three passes

Pass	What to do
1. Learn the chain	Asset → threat → vulnerability → exploit → control → risk. Every later week reuses these six words.
2. Review each week	Definitions, the comparison tables, and what each real case teaches.
3. Test yourself	Use the self-check section at the end. If you cannot answer in one or two sentences, go back to that week.

What "explain a case" means

- Name the **asset** that was harmed and which **CIA goal** it lost.
- Separate the **threat** (who or what caused harm) from the **vulnerability** (the weakness that let it work).
- Name one **control** and say whether it **prevents, detects or recovers**, and whether it lowers **likelihood** or **impact**.

> **Do not memorize dates only**
> A date with no lesson scores nothing. For every case know one sentence: *what happened* and *what it teaches*. The case table near the end has all of them.

The Course as One System

Asset	Threat	Vulnerability	Exploit	Control	Risk

Each week answers a different question:

Week	Main question	Core concepts
1	What does security protect?	Definition of security, CIA triad, five words, pause-check-report, permission, defense in depth
2	What matters, what can harm it, how serious is it?	Four asset types, threat / vulnerability / exploit / control, prevent-detect-recover, MFA, backup vs restore, risk matrix
3	How are people attacked?	Social engineering, pretext vs action, six levers, phishing family, warning signs, reporting routes
4	(no slides in the repo)	—
5	How is a device attacked?	Six malware categories, seven infection paths, endpoint controls, trusted-state recovery
6	Where does data travel, and where does security matter?	LAN/WAN, switch/AP/router, IP/port/protocol/DNS, HTTPS boundary, evil twin, firewall, segmentation

Key distinction
Week 2 gives the **framework**. Week 3 applies it to **people**, week 5 to **devices**, week 6 to **networks**. The same answer pattern works everywhere: **location or asset** → **weakness** → **control**.

Week 1 - Introduction to Computer Security

Definition

Computer security is the set of ideas, practices and controls used to reduce **unauthorized access**, **unwanted change** and **harmful disruption**.

Protect access	Protect accuracy	Protect use

Those three map onto confidentiality, integrity and availability.

What we protect

Area	Examples
Information	Grades, messages, research, identity data, payment details
Devices + services	Phones, laptops, LMS, email, cloud storage, campus Wi-Fi
People + reputation	Time, trust, privacy, safety, the ability to study or work

The CIA triad

Goal	Key question	School example	Typical failure	Typical controls
Confidentiality	Who is allowed to see it?	Only the student and authorized staff see a grade	Reading someone's email, public sharing of a private file, shoulder-surfing	Screen lock, unique password, MFA, sharing permissions
Integrity	Is it correct, complete, trustworthy?	An assignment arrives exactly as submitted	Altered grade, old file uploaded, corrupted storage	Access control, validation, logs and version history, hashes, backups
Availability	Can authorized people use it when needed?	LMS works before a deadline	DoS traffic, ransomware, power or hardware failure, bad update	Backups, redundancy, maintenance, monitoring, recovery plans

Important
Integrity and availability can fail **by accident**, with no attacker: editing the wrong row, a software bug, a power failure. An accident is still a security problem because the data is no longer trustworthy or usable.

Which CIA goal? (practise this)

Situation	Goal
A student's grades are posted publicly	C
Someone changes a submitted assignment	I
The LMS is offline before a deadline	A
A shared file is corrupted and cannot open	I (and as a result A)

Balance

- More protection can add effort: MFA improves confidentiality but adds a login step. The answer is to make protection **usable**, not to remove it.
- One control can serve several goals: a **tested backup** supports availability and can restore integrity.

- Avoid both extremes: "nothing bad has happened, so we are safe" and "everything is a disaster".

Week 1 - Five Words, Safe Response, Ethics

Five words describe a security problem

Term	Meaning	Mina's account
Asset	Something valuable	Her account, email, files, grades
Threat	A possible cause of harm	An attacker
Vulnerability	A weakness that could be used	A reused or phished password
Control	A safeguard that reduces risk	MFA, unique password, backups
Risk	The possibility and consequence of harm	Losing privacy, grades and reputation before a deadline

Case: Mina's university account

One account covers LMS, email, submissions, grades and files. One compromise hits **all three** goals: private email read (C), messages sent as her and assignment replaced (I), lockout before the deadline (A). It also harms **other people**: classmates and the professor receive the fake messages.

People, process, technology

People make choices	Process says what to do	Technology blocks, detects, records, recovers

Technology alone is never enough: a strong password system fails if a person gives the password away.

Pause, check, report

PAUSE: do not click, reply, install, share	CHECK: verify sender, request, address, context	REPORT: use the official channel

Warning signs: unexpected urgency, unfamiliar sender, mismatched address, request for credentials. **Pressure is a signal to slow down.** Verify through a route **you** choose (open the official site yourself), never through the link in the message.

Permission defines ethical practice

Authorized practice has a clear **owner, scope, time, method and purpose**.

Authorized	Unauthorized
Scanning an instructor-provided lab machine during the scheduled exercise, using approved steps, submitting findings privately	Scanning a university server because it is reachable, then sharing what you found

Do not confuse

"Reachable" does not mean "allowed". The scan can be technically identical; **permission, scope and how results are handled** make the difference. When scope is unclear: **stop and ask**.

Defense in depth

Several layers: attentive users, strong authentication, device protection, network controls, monitoring, backups. **Any single control can fail**, so another layer catches what one misses.

Week 2 - Assets

Four asset types

Type	Meaning	Examples	Real case
Information	Useful facts we depend on	Patient records, grades, passwords	Equifax 2017
Devices	Physical tools that support work	Laptops, phones, hospital computers	NHS, May 2017
People	Knowledge, work and wellbeing	Staff, patients, students	NHS, May 2017
Services	Functions people rely on	Online exams, messaging, payments	Facebook, 4 Oct 2021

An asset list is only the beginning

Question	Meaning	Example
Value	Why does it matter?	Two identical computers: one shows a public notice, one supports urgent care. **Same price, different value.**
Dependency	What does it need to work?	Facebook 2021: a network setting change cut communication between data centers. The computers were fine but the services failed.
Protection goals	What must be preserved?	Use C, I and A.

Key idea
Value comes from what the asset **does**, not what it costs. Services that share one dependency **fail together**.

What can an asset lose?

Goal	Real case
Confidentiality	Equifax: private data was exposed
Integrity	GitLab: database data was deleted
Availability	NHS: healthcare work was disrupted

Week 2 - Harm and Protection

Four roles

Term	Meaning	Equifax 2017
Threat	A possible cause of harm	The attackers
Vulnerability	A weakness	The unfixed software flaw
Exploit	A way to use a weakness	Using the flaw to get in
Control	A protective measure	A **verified** patch

Do not confuse
The attackers are the threat. The flaw is the vulnerability. Using the flaw is the exploit. **Vulnerability = the flaw. Exploit = its use.**

Three kinds of threat (M / A / P)

Malicious	Accidental	Physical
Deliberate harm	Unintended harm	Physical damage
WannaCry 2017: ransomware disrupted NHS services	GitLab 2017: database data deleted by mistake	Kakao 2022: data-center fire

A threat does not have to be a person with bad intentions.

Incident vs risk

Incident	Risk
Something **has happened** that harms or threatens security	**Possible future** harm, judged by chance and seriousness

Controls: prevent, detect, recover

PREVENT: fix a known flaw	DETECT: check records of unusual access	RECOVER: restore from a tested copy

- **Patch** = a software update that fixes a problem. Equifax lesson: a patch instruction is not enough; the patch must be **applied and checked** on the real system.
- **Backup** = make a separate copy. **Restore** = use the copy to recover. GitLab lesson: **recovery must be tested**. A backup you never restored from is an assumption.

Account controls

Term	Meaning
Phishing	A deceptive message or website that tries to trick someone
Credentials	Evidence used to prove identity, such as login details
Authentication	The process of checking a claimed identity
MFA	More than one **kind** of identity evidence: something you **know** (password) + something you **have** (security key)

Uber, Sep 2022	Cloudflare, 2022
After repeated requests, a contractor **approved** a login	Staff entered details on a fake site, but **security keys blocked** the login
Control depended on a person's judgment under pressure	**Origin-bound** key checks which website is asking, automatically
Attacker got in	Stolen login details were not enough

Week 2 - Assessing Risk

Risk = possible harm judged by **likelihood** and **impact**.

Before you rate anything

Define	Ask	Example
Event	What harmful event do we mean?	An attacker gains account access
Period	Over what period?	Before next month's exam
Current controls	What protection works now?	Security keys block one route

Separate **facts** (supported by evidence), **assumptions** (accepted for the exercise) and **unknowns** (missing evidence).

The two scales

Score	Likelihood	Impact
1	**Unlikely**: not expected under stated conditions	**Minor**: small disruption, routine recovery
2	**Possible**: a believable route to harm exists	**Serious**: important work or privacy affected, extra help needed
3	**Likely**: expected, or strongly supported by evidence	**Severe**: essential service lost, or harm hard to reverse

Risk matrix: Score = Likelihood × Impact

Likelihood ↓ / Impact →	1 Minor	2 Serious	3 Severe
3 Likely	3 Medium	6 High	9 High
2 Possible	2 Low	4 Medium	6 High
1 Unlikely	1 Low	2 Low	3 Medium

High-yield: the bands

1–2 = Low, 3–4 = Medium, 6–9 = High. The score 5 cannot occur. Always write the working: "L = 2 (possible) × I = 3 (severe) = 6, High".

Worked examples from the slides

Scenario	L	I	Score	Level
WannaCry-like disruption, essential care stops	3	3	9	High
GitLab-like deletion, but a recent restore works	2	1	2	Low
Cloudflare-like route with strong key protection	1	3	3	Medium
Update failure interrupts essential work	2	3	6	High

What each control changes

Control	Reduces	Because
Verified **patch**	**Likelihood**	That exploitation route is closed (other weaknesses may remain)
Tested **restore**	**Impact**	Less is lost after the event (some harm can remain)

The week 2 assignment pattern (exam in 30 days)

A public exam system has an unfixed flaw and an untested backup. Four sentences: *the asset is … it matters because … / the threat could use …, which is the weakness / likelihood is … and impact is … because … / our control is …; it reduces …*. Model answer: L 3 × I 3 = 9 High; **patch and verify** (likelihood) and **test the restore** (impact).

Week 3 - Social Engineering

Definition

Social engineering = deception that influences a person's action. It hacks the **human factor** (attention, emotion, habits, working situation), not the computer code. It can arrive by phone, text, email, social media or in person.

Pretext vs requested action

Pretext	Requested action
The situation the attacker invents	What the attacker wants you to do
"Your scholarship payment failed."	Send a password, approve a login, transfer money

> **Key idea**
> The story may be believable. **The action is what causes harm**, so judge the action.

Why it works

Trust: we cooperate with familiar people	Pressure: fear or a deadline narrows attention	Attention: a busy moment interrupts checking

Six levers (plus secrecy)

Lever	Sounds like
Authority	"I am from the court."
Fear	"Your account is involved in a crime."
Urgency	"You have ten minutes."
Familiarity	"It's me. I changed my number."
Reward	"You won a tuition grant."
Curiosity	"Someone posted your photo."
(Secrecy)	"Keep this private until approval." It removes the friend who would say "that looks odd".

In person

Baiting	Tailgating
An attractive item or offer invites a risky action	Someone follows an authorized person into a restricted area
Unknown USB labelled "Exam answers"	A visitor asks you to hold a secure door
Do not plug it in; hand it to IT	Stay polite: point to reception or call the host

Uber: MFA fatigue

Attacker has the password	Repeated approval prompts, first rejected	One accepted prompt gives access

Nothing about the request changed; the **person's state** changed. A better control than "be more careful": phishing-resistant MFA (origin-bound key), limits and alerts on repeated prompts, and a rule that **a prompt you did not start is reported, never approved**.

Verified contact vs claimed identity

Verified contact	Claimed identity
You reach the person through a route you already trust, and confirm the exact request	The message *says* "university", "bank", "court" or "your friend"
	Names, logos and profile pictures can be copied

Week 3 - Phishing and Impersonation

Phishing = deceptive contact that seeks information or an unsafe action.

Five labels, five different features

Term	Feature it describes	Meaning
Phishing	Deception	The broad category
Spear phishing	**Targeting**	Personalized to a person, role or group
Smishing	**Text** channel	Phishing by SMS or text
Vishing	**Voice** channel	Phishing by call
Impersonation	**Claimed identity**	Pretending to be someone else

High-yield

Each label describes a different feature, so **one attack can carry several labels**.

Contact	Labels
Generic email requests a new sign-in	Broad phishing
SMS uses your project name and a fake tutor identity	Smishing + spear phishing + impersonation
A caller posing as a bank asks for a code	Vishing + impersonation
A copied friend profile requests urgent money	Impersonation

Cases

Case	What happened	Lesson
Korean "daughter" call (reported Jul 2018)	Caller posed as a man's daughter, claimed an accident, asked for 20 million won. The real daughter was at home; no payment.	Fear + urgency + familiarity. Control: call back on a **known number**, or a family code word.
Twilio (Aug 2022)	Texts impersonated internal IT; staff used fake sign-in pages	A familiar name does not prove the destination. One account exposed **other people's** data, so limit what an account can reach.
Changwon District Court warning (8 Jul 2025)	"Court mail was returned": use a link, give information, install an app	Authority + unfamiliar legal language adds pressure. Find the court's number yourself.
Hong Kong deepfake meeting (reported 2024)	Fake CFO email, then a fabricated video meeting; about **HK$200 million** transferred	**Recognition is not authorization.**

Authorization vs recognition

Authorization	Recognition
"This request passed our agreed verification and approval process."	"That looks like my manager."

A **deepfake** uses AI to imitate a person's appearance or voice. Seeing a face raises confidence, but a sensitive request still needs an independent check.

Week 3 - Warning Signs and Reporting

U + P + S

Unexpected: new contact, unusual login	Pressure: deadline, threat, secrecy	Sensitive action: credentials, money, device access

One sign alone may be innocent. All three together is the pattern to stop on. Requests that deserve a pause: a **password or code**, a **payment change**, **device access** (install an app, enable remote access).

Read the full hostname

```
https://login.campus.example/student           real: belongs to campus.example
https://campus.example.verify-now.test/student  fake: belongs to verify-now.test
```

A hostname is read **from the right**. Open the known portal yourself.

Appearance is limited evidence

Stronger checks	Easy to imitate
An expected task and a known contact route	Logo, display name, fluent grammar, familiar layout
Confirming the exact request through the official process	**HTTPS**: it also protects a connection to a deceptive site

A QR code is a hidden link. For an urgent tuition QR notice the strongest action is **open the usual portal**, not "scan now" and not "ask the sender".

Pause, verify, report

PAUSE: stop the action, end the pressure	VERIFY: separate known route, confirm the exact request	REPORT: official channel, explain what happened

A verification route is **independent** when you obtained it **separately** from the suspicious contact.

Reporting routes in South Korea

Situation	Route	Purpose
Money sent, or active voice-phishing fraud	**112** and your **bank's official contact**	Report urgently and request **payment suspension**
Suspicious SMS or possibly malicious app	**118 KISA**	Technical advice about the message or device
Campus account or university message	Known **university IT** office	Report and arrange an account response

If you already acted

What happened	Immediate response
Opened a link only	Close it, report it, mention any downloads or permissions
Entered credentials or approved a login	Reset access and end sessions through a trusted route; contact IT
Installed an app or allowed remote access	Stop using the device for sensitive tasks; get help **from another device**
Sent money	112 and the bank **immediately**; request payment suspension

Respond supportively ("Thank you for telling me"). **Shame makes people hide information and delay help.**

Week 5 - Malware Categories

Malware = software intentionally designed to cause harm: steal information, disrupt service, spy, gain control, or enable later attacks.

> **Key idea**
> Malware is identified by its **purpose and actions**, not by its name, appearance or symptoms. Some malware hides quietly, and **a symptom alone does not prove infection**.

Six categories

Category	Slide example	How it behaves	What helps
Virus	Sality	Activates when an infected file or program is **executed**; copies itself into other content	Do not share or run untrusted files
Worm	Conficker	**Propagates across networks** by exploiting weaknesses; no user needs to open a file	Fast patching, segmentation
Trojan	SocGholish (fake browser update)	**Disguised** as a game, update, document or pirated app	Trusted sources only; a filename proves nothing
Ransomware	WannaCry	**Encrypts or blocks** systems, may also steal data	Backups; **payment does not guarantee recovery**
Spyware	Spyware.KeyLogger	**Collects information**; a keylogger captures keystrokes	Monitoring, endpoint protection (often no visible sign)
Bot	Neutrino Bot	Lets an **external operator** use the device for spam, fraud, attacks	Monitoring for hidden activity

> **Do not confuse**
> A **virus needs a user action** (running or sharing an infected file). A **worm spreads by itself** through reachable, unpatched systems. WannaCry was **ransomware that also spread as a worm**.

Name the type (practise this)

Observation	Most likely type
Infected program shared on USB; other executables become infected	Virus
Several unpatched computers infected without users opening a file	Worm
A "free game" installs an unwanted remote-control program	Trojan
Course files no longer open; a screen demands payment	Ransomware
A program secretly records what a student types	Spyware / keylogger
A device sends attack traffic when a remote operator says so	Bot
A laptop is slow, but no scan, process or network evidence exists	**Unknown, possibly not malware**
A fake update steals information and downloads ransomware	Trojan + spyware + ransomware

Evidence that separates malware from an ordinary fault: running processes, network connections, recent changes, scan results, startup and configuration changes. A computer **can** be infected with no warning window; only ransomware wants to be seen.

Week 5 - Infection Paths

Malware depends on unsafe content, weak configuration, excessive privilege or unpatched software.

Seven paths and their controls

Path	How it works	Control
Malicious messages	Attachments, links, fake login pages	Verify unexpected content; filtering; user reporting
Untrusted downloads	Pirated software, unofficial stores, fake updates	Approved sources; check publisher and update path
Unpatched vulnerabilities	Known flaws exploited when updates are delayed	Prioritize internet-facing and critical systems; replace unsupported software
Excess privilege	Admin access lets malware change more and disable controls	**Standard accounts** for daily work
Removable media	Unknown or infected USB devices	Media policy; do not connect found devices
Weak configuration	Auto-run, exposed services, default credentials, disabled protections	Secure baselines; monitor configuration changes
Supply chain	Legitimate updates or vendors are compromised	Trusted signed sources; track dependencies and supplier notices

An attachment from a **known contact** can still be dangerous: the account may be compromised, the address spoofed, or the contact forwarding it without knowing.

The infection story structure

Initial contact	Execution	Privilege	Persistence	Impact	Recovery

Story	Contact	Execution	Persistence	Control
Email attachment	User receives an infected document	User **opens** it	Startup entry or scheduled task	Filtering, verify content, standard accounts
Fake software / update	User downloads a fake game or update	User **runs** the installer	Installed program, auto-start	Approved sources, restrict installation
Network worm	Vulnerable computer is **reachable**	Worm exploits an unpatched service, **no user action**	Service or system component	Patch quickly, segmentation
Infected USB	User connects the USB	User **runs** the infected file	Copies to other files and media	Media policy, disable auto-run

High-yield

This six-stage path was the week 5 assignment. Be ready to write one story stage by stage, with a control placed against **each stage**, not all at the end.

Week 5 - Protect and Respond at the Endpoint

Four endpoint controls

Control	Detail to remember
Updates	OS, browsers, applications, **firmware**, security tools. Automatic where appropriate; **track devices that cannot update**.
Endpoint protection	Blocks known threats, flags suspicious files and processes. Detection quality changes over time; **alerts need investigation**.
Secure configuration	Disable unnecessary services, restrict installation, lock screens, encrypt storage. **Defaults prioritize convenience.** Configuration should be repeatable and reviewable.
Backups	**Protected, separated and tested.** A **connected backup may also be encrypted** by ransomware.

Least privilege limits what malicious code can change or access.

If you suspect infection

Stop sensitive work	Avoid unnecessary changes	Report quickly

- **Do not keep experimenting** on the device.
- **Follow guidance before disconnecting critical systems**: staff may need to preserve evidence.

Recovery = a trusted state

Clean system	Weakness fixed	Credentials safe	Data restored	Monitored afterwards

Trap

Restoring files alone may restore the weakness. Fix the cause, and document the cause and the improvement.

Why antivirus is not enough

No tool detects every threat. Protection also depends on configuration, updates, privilege, backups and users. This is week 1's **defense in depth**.

Six questions before recommending anything

Symptom	Infection path	Privilege level	Affected data	Safe first action	Recovery evidence

Library computer activity (model answer)

A fake update was installed, the browser opens unknown pages, and a student used email afterwards. **Behavior:** Trojan from the fake update, possible spyware. **Containment:** stop sensitive use, do not experiment, report to university IT. **Affected accounts:** anything used **after** the installation, starting with email; reset from a **different trusted device**. **Prevention:** restrict installation, standard accounts, approved sources, monitored endpoint protection.

Week 6 - Follow the Network Path

A **network** connects devices so they can exchange data.

Laptop	Access point	Switch	Router / firewall	Internet routers	LMS server

The shared printer hangs off the switch. In the labelled diagram: **A = access point, B = switch, C = Internet routers, D = router / firewall**.

LAN, WAN, Internet

Term	Meaning	Example
LAN	Network in a limited local area	Computers and a printer in one lab
WAN	Links networks over larger distances	A university links two distant sites
Internet	Many connected networks worldwide	Campus reaches a cloud LMS

Endpoint = a device at an end of a communication. **Client** starts a request; **server** provides a service.

The devices in between

Device	What it does	Security concern
Switch	Forwards Ethernet **frames inside a LAN**; learns **MAC addresses**	An unused cable socket can be an entry point
Access point	Lets **wireless** devices join a network	Who can connect, and what can they reach?
Router	Forwards **IP packets between networks**; the **default gateway** is the next router	Settings can be changed by an attacker

Key distinction

Switches connect local devices. Routers connect networks. A home "Wi-Fi router" is router + AP + switch in one box.

Local vs remote

Path	Route
Print job	Laptop → AP → switch → printer. **Stays local.**
LMS request	Laptop → AP → switch → router/firewall → Internet routers → server

If the Internet connection fails, the local printer **still works**. Wi-Fi connects you to a network, which is not the same as the Internet.

Where a risk can begin

Location	What could go wrong	Control
Device	Malware uses the device's connections	Updates and endpoint protection
Wi-Fi or cable socket	An unauthorized device joins	Network access control
Router settings	An attacker changes settings	Protected admin access and updates

WannaCry from the network side: it spread to **reachable, unpatched** Windows systems through a vulnerable **SMB** (file sharing) service. Reduce spread by installing the fix and restricting unnecessary SMB traffic.

Week 6 - Read a Network Connection

Four terms, four questions

Question	Term	Classroom example
Where is it?	**IP address**	Laptop 192.168.10.24, gateway 192.168.10.1, LMS 198.51.100.20
Which service?	**Port**	443 = HTTPS, 22 = SSH
Which rules?	**Protocol**	IP, TCP, TLS, HTTP
What address matches the name?	**DNS**	lms.example.com → 198.51.100.20

- An IP address identifies a **network interface**. **It does not prove a person's identity.**
- **NAT** (Network Address Translation) changes address information as traffic passes, so many **private** addresses share one **public** address (203.0.113.5).

Protocols

Protocol	Plain English
IP	Delivers packets between networks
TCP	Delivers data reliably and in order
TLS	Protects a connection's data: encryption and certificate checks
HTTP	Defines web requests and responses

HTTPS = HTTP protected by TLS. A **certificate** is digital evidence used to check the server's identity. **Encryption** makes content unreadable without the right keys.

DNS and the URL

Device asks: lms.example.com?	Resolver answers: 198.51.100.20	Device connects to 198.51.100.20

URL part	Meaning
`https`	Use HTTP protected by TLS
`lms.example.com`	Domain name; DNS finds its address
`/course/6`	Path to a resource
(443)	Default HTTPS port when none is written

The HTTPS protection boundary

Question	What HTTPS tells us
Can a nearby observer read the page content?	**Normally no**, when TLS works correctly
Is every website using HTTPS honest?	**No.** A phishing site can also use HTTPS
Can malware read data on my own device?	**Yes.** Endpoint security still matters
Is all network information hidden?	**No.** IP addresses and traffic timing may remain visible

High-yield

HTTPS protects **content in transit between the TLS endpoints**, and nothing else. In browser DevTools a row is a **resource request, not one packet**, and the browser shows clear content because it **is** one endpoint.

Week 6 - Locate and Reduce Risk

A useful explanation names the **location**, the **weakness** and the **control**.

Five locations

Device: endpoint protection, updates	Access: authenticated Wi-Fi	Boundary: firewall rules	Data in transit: TLS	Server: permissions

Named risks

Risk	Where	What it is	Response
Evil twin	Access	A fake access point imitating a trusted Wi-Fi name. An SSID proves only the advertised name.	Verify the network, follow approved setup, **stop at certificate warnings**
DNS spoofing	DNS	False DNS information redirects traffic	It **does not automatically defeat TLS**: the fake server still needs a valid certificate for that name
On-path attacker	In transit	Someone positioned to intercept or influence traffic	HTTP is exposed; correct HTTPS protects content
Too much exposure	Service	A server reachable in more ways than needed	Firewall: allow 443 to the public, deny public SSH 22, allow an approved admin path
Excessive access	Server	Encrypted request, but the user gets more than they should	Authentication and authorization

Firewall = software or hardware that allows or blocks traffic using rules.

Authentication vs authorization

Authentication	Authorization
Who are you? Checks **identity**	*What may you do?* Checks **permissions**

Segmentation

Divides a network into groups and controls communication between them. Guest devices → staff records server: **deny**. Guest devices → Internet: **allow**. It limits unnecessary paths and an incident's **spread**.

Small office activity (model answer)

Risk	Control	Why it helps
Guests can reach staff systems	Separate guest access, enforce traffic rules	Limits access to private services
Administration is public	Allow only an approved admin path	Reduces unnecessary exposure
Printer firmware is old	Update it, or restrict it until replaced	Reduces exploitable weaknesses

The café certificate warning

A warning does not say **where** the problem is. It could be the **device** (wrong clock, malware, intercepting software), the **network** (fake hotspot), or the **website** (expired or wrong certificate). Name the location first.

Every Real Case in One Table

Case	When	What happened	What it teaches	Week
WannaCry / NHS	May 2017	Ransomware disrupted hospital services; spread as a worm through unpatched SMB	Malicious threat, availability, patching, reachability	2, 5, 6
Equifax	2017	Attackers used an unfixed software flaw to reach personal data	Vulnerability vs exploit; a patch must be applied **and verified**; confidentiality	2
GitLab	2017	Database data deleted by mistake; recovery was hard	Accidental threat; integrity; backup is not restore	2
Facebook	4 Oct 2021	Network setting change cut data centers off; services down	Dependency; availability	2
Kakao	2022	Data-center fire disrupted services	Physical threat	2
Uber	Sep 2022	Contractor approved a login after repeated prompts	MFA fatigue; controls that rely on judgment under pressure are weak	2, 3
Cloudflare	2022	Staff typed details into a fake site; security keys blocked the login	Origin-bound, phishing-resistant MFA	2
Korean "daughter" call	Jul 2018	Vishing with family impersonation, 20 million won requested	Emotion blocks the pause; verify by a known number	3
Twilio	Aug 2022	Smishing impersonating internal IT; fake sign-in pages	A familiar name is not the destination; limit an account's reach	3
Changwon District Court	8 Jul 2025	Fake "returned court mail" calls and texts	Authority + unfamiliar legal language	3
Hong Kong deepfake	2024	Fake CFO email and fabricated video meeting, about HK$200M	Recognition is not authorization	3
Sality, Conficker, SocGholish, Spyware.KeyLogger, Neutrino Bot	—	Slide examples of virus, worm, Trojan, spyware, bot	One example per malware category	5

Commonly Confused Concepts

Do not confuse	Correct distinction
Threat vs vulnerability	Threat = what could cause harm (the attackers). Vulnerability = the weakness that lets it work (the flaw).
Vulnerability vs exploit	The flaw vs the **use** of the flaw.
Incident vs risk	Has happened vs possible future harm.
Likelihood vs impact	Chance of the event vs seriousness of the consequences.
Patch vs restore	A patch lowers **likelihood**; a tested restore lowers **impact**.
Backup vs restore	Making a copy vs using it to recover. Only a **tested** restore counts.
Confidentiality vs integrity	Who may **see** it vs whether it is **correct**.
Integrity vs availability	Correct and complete vs usable when needed. A corrupted file can lose both.
Authentication vs authorization	Who are you vs what may you do.
Recognition vs authorization	"Looks like my manager" vs "passed our approval process".
Pretext vs requested action	The invented story vs what they want you to do.
Phishing vs spear phishing	General message to many vs tailored to a person or group.
Smishing vs vishing	Text channel vs voice channel.
Baiting vs tailgating	Tempting item or offer vs following someone through a secure door.
Virus vs worm	Needs a user to run or share it vs spreads by itself across the network.
Trojan vs virus	Trojan is **disguised** as something useful; a virus **infects other files**.
Spyware vs bot	Collects your information vs lets an outside operator use your device.
Symptom vs evidence	A slow laptop vs scan, process, network or configuration findings.
Switch vs router	Frames inside a LAN vs packets between networks.
Wi-Fi vs Internet	A local network connection vs the worldwide set of networks.
IP address vs port	Which destination vs which service there.
IP address vs identity	An address identifies an interface; it does not prove who the person is.
HTTP vs HTTPS	No protection in transit vs TLS-protected content.
HTTPS vs "safe site"	Encrypted connection vs honest owner. A phishing site can use HTTPS.
DNS spoofing vs broken TLS	A false address redirects traffic, but the certificate check still stands.
Evil twin vs on-path attacker	A fake access point vs anyone positioned to intercept traffic.
Firewall vs segmentation	Rules that allow or block traffic vs dividing the network into controlled groups.
Authorized vs reachable	Permission with owner, scope, time, method, purpose vs "I could connect to it".

How to Answer Scenario Questions

1. "Which CIA goal?"

Seen by the wrong person = C	Changed or wrong = I	Cannot be used when needed = A

Name the goal, then justify with the definition in half a sentence. If two apply, say which is **most** affected and why.

2. "Label this incident"

Asset	Threat	Vulnerability	Exploit	Control

Model: *Equifax. Asset: personal information. Threat: attackers. Vulnerability: the unfixed software flaw. Exploit: using it to gain access. Control: a verified patch.*

3. "Rate the risk"

Define event, period, current controls	Likelihood 1–3 with reason	Impact 1–3 with reason	Multiply	Band	Control and what it lowers

4. "Is this message safe?"

Name the pretext	Name the levers	Name the requested action	Give the independent check	Give the reporting route

5. "What malware is this?" / "Trace the infection"

Behavior	Category	What evidence would confirm it	Path: contact, execution, privilege, persistence, impact	Control per stage	Trusted-state recovery

6. "Where is the network risk?"

Location	Weakness	Control that fits that location

Model: *Wi-Fi access. Unknown network operator (evil twin). Verify the approved network and stop at certificate warnings.* Pick risks at **different** locations when asked for two.

Self-Check Questions

Try to answer without looking back. These are review questions, not copies of the real midterm.

1. Define computer security in one sentence.
2. What do C, I and A stand for, and what question does each ask?
3. Which CIA goal is most affected when a grade is changed without permission?
4. Why can an accidental error still be a security problem?
5. Name the five words that describe a security problem.
6. What five things define authorized security practice?
7. Why do secure systems use several layers of defense?
8. Name the four asset types with one example each.
9. Two computers cost the same. Why might one be more valuable?
10. In Equifax, what was the threat, the vulnerability and the exploit?
11. Give one example each of a malicious, an accidental and a physical threat.
12. What is the difference between an incident and a risk?
13. Name the three things a control can do, with an example each.
14. Compare what happened at Uber and at Cloudflare in 2022.
15. Why are backup and restore different actions?
16. Likelihood 2 and impact 3: what is the score and the level?
17. Does a patch mainly reduce likelihood or impact? What about a tested restore?
18. What three things must you define before rating a risk?
19. Define social engineering.
20. What is the difference between a pretext and a requested action?
21. Name the six levers attackers use.
22. A caller posing as a bank asks for a code. Which labels apply?
23. Why is "recognition is not authorization" the lesson of the Hong Kong case?
24. Which address is fake and why: `login.campus.example` or `campus.example.verify-now.test`?
25. Which Korean reporting route fits: (a) money already sent, (b) a suspicious SMS, (c) a strange campus-account email?
26. What makes a verification route independent?
27. How is a worm different from a virus?
28. What is a Trojan? Give the slide example.
29. A laptop is slow and nothing else is known. What malware is it?
30. Name four of the seven infection paths.
31. Write the six stages of an infection story.
32. Why may a connected backup be useless against ransomware?
33. What five conditions make recovery complete?
34. Why is antivirus not enough?
35. What is the difference between a switch and a router?
36. The Internet is down. Can you still print to the lab printer? Why?
37. What do an IP address, a port, a protocol and DNS each answer?
38. Name two things HTTPS does **not** protect against.
39. What is an evil twin, and what should you do at a certificate warning?
40. Does DNS spoofing defeat TLS? Explain.
41. Authentication vs authorization?
42. How does segmentation reduce harm?

Self-Check Answers

1. The ideas, practices and controls used to reduce unauthorized access, unwanted change and harmful disruption.
2. Confidentiality: who may see it? Integrity: is it correct, complete, trustworthy? Availability: can authorized people use it when needed?
3. Integrity: the information was changed improperly.
4. It can make data incorrect, expose information or stop a service even with no attacker.
5. Asset, threat, vulnerability, control, risk.
6. A clear owner, scope, time, method and purpose.
7. Any single control can fail; other layers reduce the chance that one mistake causes major harm.
8. Information (grades), devices (laptops), people (staff), services (online exam).
9. Value comes from what the asset does: one may support an urgent service, so interruption costs more.
10. Threat: the attackers. Vulnerability: the unfixed software flaw. Exploit: using that flaw to gain access.
11. Malicious: WannaCry. Accidental: GitLab deletion. Physical: Kakao data-center fire.
12. An incident has happened; a risk is possible future harm judged by likelihood and impact.
13. Prevent (fix a known flaw), detect (check records of unusual access), recover (restore from a tested copy).
14. Uber: a person approved a login after repeated prompts, so the attacker got in. Cloudflare: staff typed details into a fake site, but origin-bound security keys refused to work there.
15. Backup makes a copy; restore uses it to recover. An untested backup may not restore (GitLab).
16. 2 × 3 = 6, High.
17. A patch reduces likelihood. A tested restore reduces impact.
18. The event, the period and the current controls.
19. Deception that influences a person's action.
20. The pretext is the invented situation; the requested action is what the attacker wants you to do, and it is the action that causes harm.
21. Authority, fear, urgency, familiarity, reward, curiosity.
22. Vishing + impersonation.
23. The employee recognized faces in a video meeting, but the request never passed an independent approval process.
24. The second: reading from the right, it belongs to `verify-now.test`.
25. (a) 112 and the bank, request payment suspension. (b) 118 KISA. (c) University IT.
26. You obtained it separately from the suspicious contact.
27. A worm propagates through networks by itself; a virus depends on a user executing or sharing an infected file.
28. Malware disguised as legitimate or useful software. SocGholish, a fake browser update.
29. Unknown. A symptom is not evidence; you need scan, process, network or configuration findings.
30. Any four of: malicious messages, untrusted downloads, unpatched vulnerabilities, excess privilege, removable media, weak configuration, supply chain.
31. Initial contact, execution, privilege, persistence, impact, recovery.
32. Ransomware can encrypt it too. Backups must be protected, separated and tested.
33. Clean system, weakness fixed, credentials safe, data restored, monitored afterwards.
34. No tool detects every threat; protection also needs configuration, updates, least privilege, backups and users.
35. A switch forwards frames inside a LAN; a router forwards packets between networks.
36. Yes. The print path (laptop → AP → switch → printer) never leaves the LAN.
37. IP: where is it. Port: which service. Protocol: which rules. DNS: which address matches the name.
38. Any two of: a dishonest site owner (phishing sites use HTTPS), malware on your own device, visibility of IP addresses and traffic timing.

39. A fake access point imitating a trusted Wi-Fi name. Stop; do not click through the warning.
40. Not by itself. It can redirect traffic, but the fake server must still present a valid certificate for the name.
41. Authentication checks who you are; authorization checks what you may do.
42. It limits unnecessary communication between groups, so an incident cannot spread freely.

Final 15-Minute Revision Checklist

You should be able to...	Check
Define computer security and the three CIA goals, each with a school example.	☐
Classify a scenario as C, I or A and justify it.	☐
Define asset, threat, vulnerability, exploit, control, risk, incident.	☐
Explain pause-check-report, permission (owner, scope, time, method, purpose) and defense in depth.	☐
List the four asset types and explain value and dependency.	☐
Label the Equifax incident.	☐
Give malicious, accidental and physical threat examples.	☐
Explain prevent / detect / recover, patch, backup vs restore.	☐
Compare Uber and Cloudflare; explain MFA and origin-bound keys.	☐
Draw the 3 × 3 risk matrix with the bands 1–2, 3–4, 6–9.	☐
Say which control lowers likelihood and which lowers impact.	☐
Define social engineering, pretext, requested action and the six levers.	☐
Apply the five phishing labels to a contact, with multiple labels.	☐
Tell the Korean call, Twilio, Changwon and Hong Kong cases with one lesson each.	☐
Read a hostname from the right; explain why HTTPS and logos prove little.	☐
Give the Korean reporting routes (112 + bank, 118 KISA, university IT) and the "already acted" responses.	☐
Define the six malware categories with the slide example for each.	☐
Explain virus vs worm and why WannaCry is both ransomware and worm.	☐
List the seven infection paths with a control for each.	☐
Write an infection story: contact → execution → privilege → persistence → impact → recovery.	☐
List the endpoint controls and the five trusted-state conditions.	☐
Trace laptop → AP → switch → router/firewall → Internet → server; label A–D.	☐
Explain LAN/WAN/Internet, switch/AP/router, IP/port/protocol/DNS, NAT.	☐
Reproduce the HTTPS protection boundary table.	☐
Explain evil twin, DNS spoofing, on-path attacker, firewall, authentication vs authorization, segmentation.	☐
Explain any network risk as location → weakness → control.	☐

Best final strategy

Do not memorize isolated definitions only. Pick one case (WannaCry is the best: it appears in weeks 2, 5 and 6) and practise telling it three ways: as a risk, as malware, and as a network path.

End of Midterm Study Guide